

Encyclopedia of the Actuary

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Preface

Hello!

1. Mathematics and Programming Toolbox

Hi

2. Probability and statistics

This chapter is primarily based on [D. D. Wackerly, W. Mendenhall, and R. L. Scheaffer \[1\]](#).

2.1 Probability

2.1.1 Foundations

An **experiment** is the process by which an observation is made.

When an experiment is performed, it can result in one or more outcomes, which are called **events**.

A **simple** event is an event that cannot be decomposed. Because sets are collection of points, we associate a distinct point, called a **sample point**, with each and every simple event associated with an experiment.

The letter E with a subscript will be used to denote a simple event or the corresponding sample point.

The **sample space** associated with an experiment is the set consisting of all possible sample points. A sample space will be denoted by S .

A **discrete** sample space is one that contains either a finite or a countable number of distinct sample points.

An **event** in a discrete sample space S is a collection of sample points—that is, any subset of S .

Suppose S is a sample space associated with an experiment. To every event A in S (A is a subset of S), we assign a number, $P(A)$, called the **probability** of A , so that the following axioms hold:

Axiom 1: $P(A) \geq 0$.

Axiom 2: $P(S) = 1$.

Axiom 3: If $A_1, A_2, A_3, \dots$ form a sequence of pairwise mutually exclusive events in S (that is, $A_i \cap A_j = \emptyset$ if $i \neq j$), then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i). \quad (2.1)$$

We can easily show that Axiom 3, which is stated in terms of an infinite sequence of events, implies a similar property for a finite sequence.

Specifically, if $A_1, A_2, \dots, A_n$ are pairwise mutually exclusive events, then

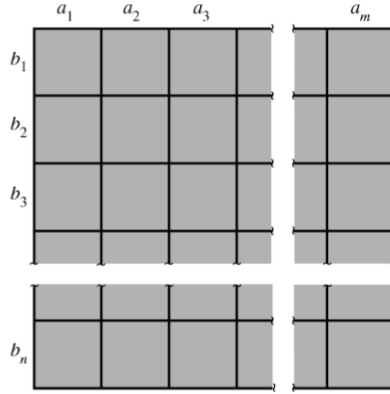
$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i). \quad (2.2)$$

If A is an event, then

$$P(A) = 1 - P(\bar{A}). \quad (2.3)$$

2.1.2 Useful results from the theory of combinatorial analysis

With m elements $a_1, a_2, \dots, a_m$ and n elements $b_1, b_2, \dots, b_n$, it is possible to form $mn = m \times n$ pairs containing one element from each group.



The mn rule can be extended to any number of sets. Given three sets of elements— $a_1, a_2, \dots, a_m; b_1, b_2, \dots, b_n$; and $c_1, c_2, \dots, c_p$ —the number of distinct triplets containing one element from each set is equal to mnp .

An **ordered arrangement** of r distinct objects is called a **permutation**. The number of ways of ordering n distinct objects taken r at a time will be designated by the symbol P_r^n . In a permutation, the order matters: arranging the same objects in a different order gives a different permutation.

$$P_r^n = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}. \quad (2.4)$$

The number of ways of **partitioning** n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects, respectively, where each object appears in exactly one group and $\sum_{i=1}^k n_i = n$, is

$$N = \binom{n}{n_1 n_2 \dots n_k} = \frac{n!}{n_1! n_2! \dots n_k!}. \quad (2.5)$$

The terms $\binom{n}{n_1 n_2 \dots n_k}$ are often called **multinomial coefficients** because they occur in the expansion of the multinomial term $y_1 + y_2 + \dots + y_k$ raised to the n th power:

$$(y_1 + y_2 + \dots + y_k)^n = \sum \binom{n}{n_1 n_2 \dots n_k} y_1^{n_1} y_2^{n_2} \dots y_k^{n_k}, \quad (2.6)$$

where this sum is taken over all $n_i = 0, 1, \dots, n$ such that $n_1 + n_2 + \dots + n_k = n$.

Additional explanation This sum literally means: go through every possible combination of exponents $n_1, n_2, \dots, n_k$ that add up to n , and create one term for each combination.

For example, consider

$$(y_1 + y_2 + y_3)^3. \quad (2.7)$$

The possible exponent combinations (n_1, n_2, n_3) must satisfy

$$n_1 + n_2 + n_3 = 3. \quad (2.8)$$

So the summation runs over

$$(3, 0, 0), (0, 3, 0), (0, 0, 3), \quad (2.9)$$

$$(2, 1, 0), (2, 0, 1), (1, 2, 0), (0, 2, 1), (1, 0, 2), (0, 1, 2), \quad (2.10)$$

and

$$(1, 1, 1). \quad (2.11)$$

Therefore,

$$(y_1 + y_2 + y_3)^3 = \sum_{\substack{n_1+n_2+n_3=3 \\ n_1, n_2, n_3 \geq 0}} \binom{3}{n_1 n_2 n_3} y_1^{n_1} y_2^{n_2} y_3^{n_3}. \quad (2.12)$$

Expanding term by term gives

$$(y_1 + y_2 + y_3)^3 = y_1^3 + y_2^3 + y_3^3 + 3y_1^2y_2 + 3y_1^2y_3 + 3y_1y_2^2 + 3y_2^2y_3 + 3y_1y_3^2 + 3y_2y_3^2 + 6y_1y_2y_3. \quad (2.13)$$

The number of **combinations** of n objects taken r at a time is the number of subsets, each of size r , that can be formed from the n objects. This number will be denoted by C_r^n or $\binom{n}{r}$.

$$C_r^n = \binom{n}{r} = \frac{n!}{r!(n-r)!}. \quad (2.14)$$

2.1.3 Conditional probability

The **conditional probability** of an event A , given that an event B has occurred, is equal to

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad (2.15)$$

provided $P(B) > 0$.

Two events A and B are said to be **independent** if any one of the following holds:

$$\begin{aligned} P(A|B) &= P(A), \\ P(B|A) &= P(B), \\ P(A \cap B) &= P(A)P(B). \end{aligned} \quad (2.16)$$

Otherwise, the events are said to be dependent.

The probability of the **union** of two events A and B is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (2.17)$$

If A and B are mutually exclusive events, $P(A \cap B) = 0$.

References

Bibliography

- [1] D. D. Wackerly, W. Mendenhall, and R. L. Scheaffer, *Mathematical Statistics with Applications*, 7th ed. Belmont, CA: Thomson Brooks/Cole, 2008.